20/08/2024

LSM

$Y = X\beta + \varepsilon$

G-M Theorem: If $a^T\beta$ is estimable then $a^T\hat{\beta}$ is the BLUE of $a^T\beta$ where $\hat{\beta}$ is any least squares estimate of β.

One can estimate $M = X\beta = \mathbb{E}Y$ and use this estimator to obtain a BLUE for $a^T\beta$

Two possible estimators of $\mu = \mathbb{E}Y$

1. Unconstrained LS estimate of μ
2. Constrained LS estimate of μ.

$(\mu = X\beta) \qquad a^T\hat{\beta} = u^T X\hat{\beta} = u^T\hat{\mu}$

Constrained LS estimate of μ

$$\hat{\mu} = \underset{\mu = X\beta \text{ for some } \beta}{\operatorname{argmin}} \|Y - \mu\|_2^2 = \underset{\mu \in \mathcal{C}(X)}{\operatorname{argmin}} \|Y - \mu\|_2^2$$

$$= P_X Y$$

$$= X\hat{\beta} \text{ where } \hat{\beta} \text{ is any LS estimate.}$$

Unconstrained LS estimate of μ

$$\tilde{\mu} = \underset{\mu}{\operatorname{argmin}} \|Y - \mu\|_2^2 = Y$$

$a^T\beta$ estimable: $\quad a \in \mathcal{R}(X) = \mathcal{C}(X^T) = \mathcal{C}(X^TX)$

$\Rightarrow \exists$ a $v \in \mathbb{R}^p$ s.t. $a = X^T(Xv) = X^Tu$

$\Rightarrow u^TY = v^TX^TY - v^TX^TX\hat{\beta} = a^T\hat{\beta}$

$$a^T\beta = u^T X^T X\beta$$

Remark: If X is full column rank, then approach I always works. (using unconstrained L.S.)

$$u^T Y = u^T P_X Y + u^T (I - P_X) Y$$
$$= u^T X\hat{\beta} + u^T \underbrace{(I - P_X)}_{\overset{\shortparallel}{0} \Leftrightarrow \text{rank}(X) = n} Y$$

If X is of full row rank

$$a = X^T X v \qquad a^T\hat{\beta} = u^T Y = v^T(X^T Y)$$

Consider the parameter $\underset{p\times n}{X^T}\,\underset{n\times 1}{\mu} = \underset{p\times 1}{\theta}$

Is $X^T Y$ the BLUE of $\theta = X^T\mu$?

$X^T\mu = X^T X\beta$ ✓

<u>Ex 1</u>: $Y_{ij} = \alpha_i + \gamma_j + \varepsilon_{ij}$ $\qquad i,j = 1,2$

$\varepsilon_{ij} \overset{iid}{\sim} (0, \sigma^2)$

$\mu_{ij} = \alpha_i + \gamma_j$

$$\underline{Y} = \begin{pmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{22} \end{pmatrix} \qquad \mu = \begin{pmatrix} \mu_{11} \\ \mu_{12} \\ \mu_{21} \\ \mu_{22} \end{pmatrix} = \begin{pmatrix} \alpha_1 + \gamma_1 \\ \alpha_1 + \gamma_2 \\ \alpha_2 + \gamma_1 \\ \alpha_2 + \gamma_2 \end{pmatrix}$$

$$\theta = X^T\mu = \begin{bmatrix} \mu_{11}+\mu_{12} \\ \mu_{21}+\mu_{22} \\ \mu_{11}+\mu_{21} \\ \mu_{12}+\mu_{22} \end{bmatrix} = \begin{bmatrix} 2\alpha_1+\gamma_1+\gamma_2 \\ 2\alpha_2+\gamma_1+\gamma_2 \\ \alpha_1+\alpha_2+2\gamma_1 \\ \alpha_1+\alpha_2+2\gamma_2 \end{bmatrix} \quad X = \begin{bmatrix} 1&0&1&0 \\ 1&0&0&1 \\ 0&1&1&0 \\ 0&1&0&1 \end{bmatrix}$$

Consider the estimable parameters: $a^T\beta$ where $a_1+a_2 = a_3+a_4$

(1) $\alpha_1 - \alpha_2 + \gamma_1 - \gamma_2 = (\alpha_1+\gamma_1) - (\alpha_2+\gamma_2)$

(2) $\alpha_1 - \alpha_2 = (\alpha_1+\gamma_1) - (\alpha_2+\gamma_1)$

We want to express these parameters as $v^T\theta$ for some v.

(1) $(\alpha_1+\gamma_1) - (\alpha_2+\gamma_2) = \mu_{11} - \mu_{22}$

$$= (\mu_{11}+\mu_{12}) - (\mu_{12}+\mu_{22})$$

$$= \theta_1 - \theta_4$$

(2) $$\alpha_1 - \alpha_2 = \frac{(2\alpha_1+\gamma_1+\gamma_2) - (2\alpha_2+\gamma_1+\gamma_2)}{2}$$

$$= \frac{\theta_1 - \theta_2}{2}$$

Can we compute $\hat{\mu}$ = constrained LS estimate of μ

$\hat{\mu} = P_{\tilde{X}} Y$

$\text{rank}(\tilde{X}) = 3 < n = 4$

$(I - P_X)$ = orthogonal projector onto $\mathcal{C}(X)^\perp$

$\downarrow$

rank 1

$$\mathcal{C}(X)^{\perp} = \text{span}\{w\} \text{ s.t. } w^T X = 0 \quad \underset{4\times 1}{w} \in \mathbb{R}^4$$

$$X = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \quad w_1 + w_2 = 0$$
$$w_3 + w_4 = 0$$
$$w_1 + w_3 = 0$$
$$w_2 + w_4 = 0$$

$$w = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \qquad P_X = I - \underbrace{\left(\frac{ww^T}{\|w\|^2}\right)}_{P_w}$$

$$= I - \frac{ww^T}{4}$$

$$X\hat{\beta} = P_X Y = Y - \tfrac{1}{4}(w^T Y) w = \underline{Y} - \left(\frac{Y_{11} - Y_{12} - Y_{21} + Y_{22}}{4}\right) \underline{w}$$

We considered one factor ANOVA model

$$Y_{ij} = \mu_i + \varepsilon_{ij} \qquad j = 1, \dots, n$$
$$= \mu + \alpha_i + \varepsilon_{ij} \qquad i = 1, \dots, r$$

$$\sum_{i=1}^{r} n_i \alpha_i = 0 \quad \left(\begin{array}{c}\text{identifiability} \\ \text{constraint}\end{array}\right)$$

$i = 1, \dots, r$ are the different "treatments"
(categorical factor with r different levels)

Ex One factor ANCOVA (analysis of covariance)
one categorical predictor (factor) with r level
one numerical predictor, say Z.

$$(Y_{ij}, Z_{ij}) \qquad \begin{array}{l} j=1,\dots,n \\ i=1,\dots,r \end{array}$$

$$Y_{ij} = \mu_i + \underset{\substack{\uparrow \\ \text{scalar}}}{\gamma} Z_{ij} + \varepsilon_{ij}$$

Our interests could be checking

$\gamma = 0$? ⓐ Is there any effect of the covariate Z?

$\mu_1 = \dots = \mu_r$? ⓑ Is there any differential effects of the schools?

ⓒ Is the covariate effect the same across schools?

$$Y_{ij} = \mu_i + \gamma_i Z_{ij} + \varepsilon_{ij}$$

$$\gamma_1 = \dots = \gamma_r$$

$$\sum_{i=1}^{r} \left(\sum_{j=1}^{n} (Y_{ij} - \mu_i - \gamma_i Z_{ij})^2 \right)$$

→ simple linear regression.

$$\underset{\sim}{Y} = \begin{bmatrix} Y_{11} \\ \vdots \\ Y_{1n_1} \\ Y_{21} \\ \vdots \\ Y_{2n_2} \\ \vdots \\ Y_{r1} \\ \vdots \\ Y_{rn_r} \end{bmatrix} = \begin{bmatrix} \underset{\sim}{1}_{n_1} & 0 & 0 & \begin{matrix} Z_{11} \\ \vdots \\ Z_{1n_1} \end{matrix} \\ 0 & \underset{\sim}{1}_{n_2} & 0 & \\ 0 & 0 & \ddots & \\ & & \underset{\sim}{1}_{n_r} & \begin{matrix} Z_{r1} \\ \vdots \\ Z_{rn_r} \end{matrix} \end{bmatrix} \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_r \\ \gamma \end{bmatrix} + \underset{\sim}{\varepsilon}$$

$$Z_{i\cdot} = \begin{pmatrix} Z_{i1} \\ \vdots \\ Z_{in_i} \end{pmatrix} \qquad \underset{(r+1)\times(r+1)}{X^T X} = \left[\begin{array}{cccc|c} n_1 & & & 0 & \sum Z_{1j} \\ & n_2 & & & \sum Z_{2j} \\ & & \ddots & & \vdots \\ 0 & & & n_r & \sum Z_{rj} \\ \hline & & & & \sum_i \sum_j Z_{ij}^2 \end{array}\right]$$

$$\begin{aligned} Y_{ij} &= \mu_i + \nu Z_{ij} + \varepsilon_{ij} \\ &= (\mu_i + \nu \overline{Z_{i\cdot}}) + \gamma \underbrace{(Z_{ij} - \overline{Z_{i\cdot}})}_{\tilde{Z}_{ij}} + \varepsilon_{ij} \end{aligned}$$

$$\left(\overline{Z_{i\cdot}} = \frac{1}{n_i} \sum_{j=1}^{n_i} Z_{ij} \right)$$

$$= \theta_i + \nu \tilde{Z}_{ij} + \varepsilon_{ij}$$

$$\hat{\theta}_i = \overline{Y}_{i\cdot} \quad \text{for } i = 1, \dots, r$$

$$\hat{\gamma} = \frac{\sum_i \sum_j \tilde{Z}_{ij} Y_{ij}}{\sum_i \sum_j \tilde{Z}_{ij}^2}$$

$$\hat{\mu}_i \stackrel{?}{=} \hat{\theta}_i - \hat{\gamma} \overline{Z_{i\cdot}}$$

$$\min_{\beta} \| Y - X\beta \|^2 \qquad \beta = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_r \\ \gamma \end{pmatrix} \qquad \beta \overset{1-1}{\longleftrightarrow} \eta$$

$$\min_{\eta} \| Y - \underbrace{\tilde{X}}_{XC^{-1}} \eta \|^2 \qquad \eta = \begin{pmatrix} \theta_1 \\ \vdots \\ \theta_r \\ \gamma \end{pmatrix}$$

$$XC^{-1} \underbrace{C\beta}_{\eta}$$